

NHTSA DSFS Meta-Analysis

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Background

Quantitative meta-analysis of the effectiveness of Dynamic Speed Feedback Signs (DSFS) in reducing driving speeds. Three sets of hypotheses are tested:

1. Activation of the signs. There are three way to carry out the test of activation: comparing before and during at the same site adjacent to the DSFS, upstream and adjacent during activation, or the difference from upstream to adjacent before to the difference from upstream to adjacent during.
2. Activation of the signs, donwstream. As above, there are muliple sets of comparisons for assessing activation effect downstream: comparing during downstream to before downstream, downstream and adjacent during activation, or a normalized difference.
3. Deactivation of the signs. Compare after and during at the same site, after and during

Activation hypothesis: H1

H1A:

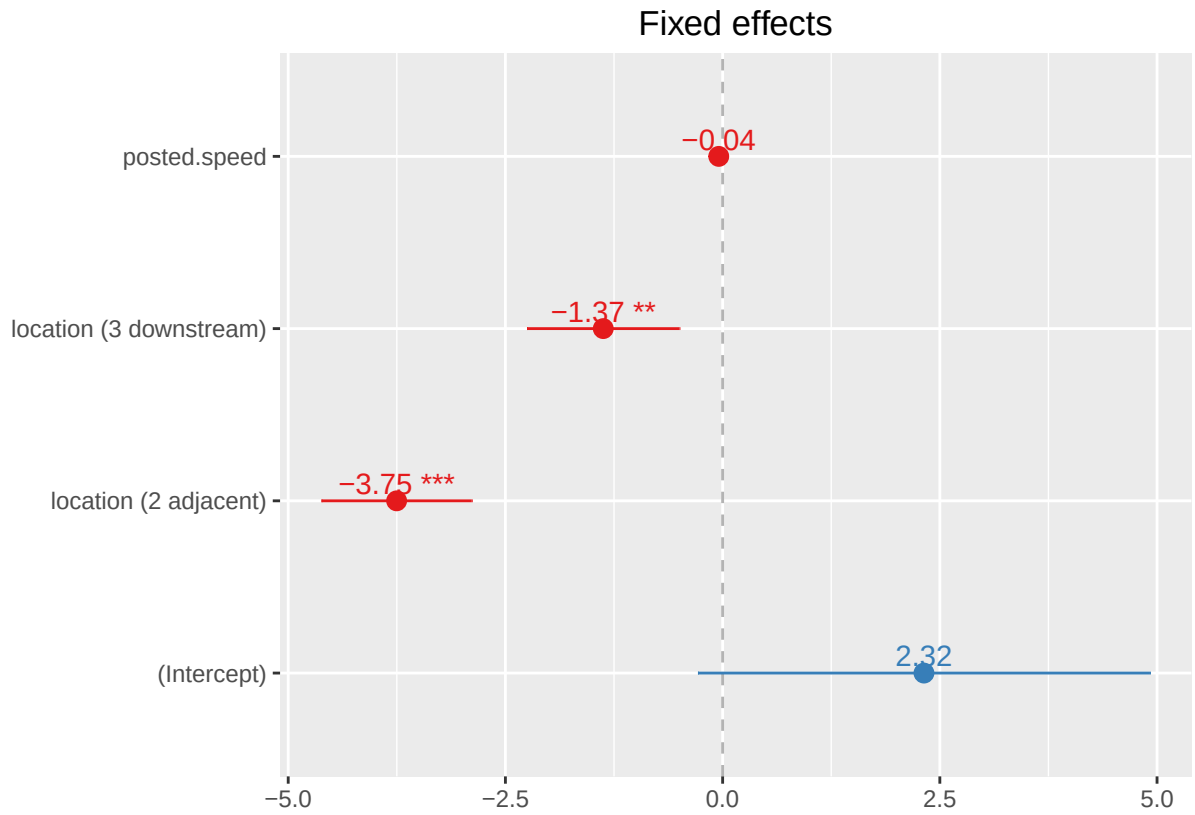
Location	Vehicle Type	Effect size y_i	Variance v_i	NA
1 upstream	Unspecified	-0.232	0.153	52
2 adjacent	Unspecified	-2.687	0.136	130
3 downstream	Unspecified	-3.072	0.195	28
1 upstream	Passenger	-0.476	0.470	59
2 adjacent	Passenger	-4.521	0.601	97
3 downstream	Passenger	-2.725	0.411	72
1 upstream	Truck	0.394	2.167	92
2 adjacent	Truck	-2.129	1.717	121
3 downstream	Truck	-2.207	0.879	74

```
## Linear mixed model fit by REML ['lmerMod']
## Formula:
## yi ~ location + posted.speed + (1 | safety.focus2) + (1 | vehicle.type2) +
##      (1 | PublicationID/StudyID/site)
## Data: h1b.md
##
## REML criterion at convergence: 2568.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
```

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## -9.1284 -0.4345  0.0338  0.5185  2.6947
##
## Random effects:
##   Groups                Name          Variance Std.Dev.
##   site:(StudyID:PublicationID) (Intercept) 0.3984   0.6312
##   StudyID:PublicationID        (Intercept) 1.5459   1.2433
##   PublicationID                (Intercept) 0.8985   0.9479
##   safety.focus2                (Intercept) 0.0000   0.0000
##   vehicle.type2                (Intercept) 0.4727   0.6876
##   Residual                     8.6562   2.9421
## Number of obs: 499, groups:
## site:(StudyID:PublicationID), 129; StudyID:PublicationID, 60; PublicationID, 25; safety.foc
##
## Fixed effects:
##               Estimate Std. Error t value
## (Intercept)      2.31734    1.32720    1.746
## location2 adjacent -3.75216    0.44123   -8.504
## location3 downstream -1.37382    0.44749   -3.070
## posted.speed      -0.04439    0.02298   -1.932
##
## Correlation of Fixed Effects:
##              (Intr) lctn2a lctn3d
## lctn2adjcnt -0.516
## lctn3dwnstr -0.331  0.616
## posted.sped -0.892  0.351  0.155
## Computing p-values via Kenward-Roger approximation. Use `p.kr = FALSE` if computation takes

```



Computing p-values via Kenward-Roger approximation. Use `p.kr = FALSE` if computation takes